Anurag Das

Saarbrücken, Germany | andas@mpi-inf.mpg.de | anurag-198.github.io

Research Profile

Final-year PhD candidate at the Max Planck Institute for Informatics, advised by Prof. Bernt Schiele. Research on failure modes and reliability of vision-language models, test-time adaptation, and semantic segmentation.

Research Experience

Max Planck Institute for Informatics *PhD Researcher, Computer Vision* Oct 2021–present

- Identified attention collapse across images as a cause of multi-image failures in VLMs. Proposed an attention-masking fix (Findings of ACL 2026).
- Co-developed ClipTTT, CLIP-guided test-time training that reduces LVLM hallucination on corrupted images.
- Developed MTA-CLIP for language-guided semantic segmentation (ECCV 2024) and a prototypical contrastive approach to weakly supervised domain adaptation (CVPR 2023).

Samsung Research Cambridge *Research Intern, Future Interaction Group* Jun–Nov 2025

- Trained 7B LLaVA-OneVision on 64 H100 GPUs with single-image, multi-image, and video instruction data.
- Analyzed model failure modes under visual corruptions and multi-image inputs, working with Brais Martinez and Adrian Bulat.

Max Planck Institute for Informatics *Master's Thesis Researcher* Oct 2020–Apr 2021

- Researched zero-label semantic segmentation under the supervision of Prof. Bernt Schiele.

Selected Publications

ACL 2026 Findings
More Images, More Problems? A Controlled Analysis of VLM Failure Modes.
A. Das, A. Bulat, A. Baldrati, I. Metaxas, B. Schiele, G. Tzimiropoulos, B. Martinez.

NeurIPS 2026 VLM4RWD Workshop
OneScene-Bench: Do Vision-Language Models Integrate Evidence Across Views?
S. Suhas, **A. Das**, B. Schiele, J. Fischer.

arXiv 2026
ClipTTT: CLIP-Guided Test-Time Training Helps LVLMs See Better.
M. Nath*, **A. Das***, J. Xie, B. Schiele.

In submission 2026
From Zeroes To Heroes: Repurposing Redundant Parameters For Efficient Adaptation of Vision Foundation Models.
S. Priyadarshi, S. Gairola, S. Agnihotri, **A. Das**, M. Keuper, B. Schiele.

TMLR 2026
Do Instance Priors Help Weakly Supervised Semantic Segmentation?
A. Das, A. Kukleva, X. Hu, Y. Asano, B. Schiele.

ECCV 2024
MTA-CLIP: Language-Guided Semantic Segmentation with Mask-Text Alignment.
A. Das, X. Hu, L. Jiang, B. Schiele.

ECCV 2024
Improving 2D Feature Representations by 3D-Aware Fine-Tuning.
Y. Yue, **A. Das**, F. Engelmann, S. Tang, J. E. Lenssen.

CVPR 2023
Weakly-Supervised Domain Adaptive Semantic Segmentation with Prototypical Contrastive Learning.
A. Das, Y. Xian, D. Dai, B. Schiele.

WACV 2023
Urban Scene Semantic Segmentation with Low-cost Coarse Annotation.
A. Das, Y. Xian, Y. He, Z. Akata, B. Schiele.

GCPR 2021
(SP)^{2}Net for Generalised Zero Label Semantic Segmentation.
A. Das, Y. Xian, Y. He, B. Schiele, Z. Akata.

*Equal contribution.

Education

PhD in Computer Vision, Max Planck Institute for Informatics 2021–present
MSc in Computer Science, Saarland University GPA: 1.4 2019–2021
BTech in Information Technology, IIIT Allahabad GPA: 9.44/10 2013–2017

Additional Experience

Saarland University, Research Intern, Modelling and Simulation Group 2019
Qualcomm, Software Engineer 2017–2019
Microsoft, Software Developer Intern 2016
WorldQuant LLC, Student Quant Consultant 2015–2016
IIT Bombay, Summer Intern 2015

Teaching and Mentoring

Master's thesis supervision, Mriganka Nath (ClipTTT) 2026
Master's thesis supervision, Luisa Danalachi (Incremental Domain Adaptation) 2023
Master's thesis supervision, Nikhil Paliwal (Domain Adaptation with Temporal Cues) 2023
Teaching Assistant, Image Processing and Computer Vision, Saarland University 2020
Teaching Assistant, Introduction to Programming, IIIT Allahabad 2015

Academic Service

Reviewer CVPR, ECCV, ICCV, NeurIPS, GCPR, TPAMI, IJCV, TMLR, IEEE-TCVST

Technical Skills

Frameworks Python, PyTorch, Hugging Face (Transformers, Accelerate, PEFT), vLLM
Infrastructure CUDA, Slurm, DeepSpeed (ZeRO), FSDP

Talks

Multi-Image Reasoning with LVLMs. *Google Research, Munich*, 2026
Weakly Supervised Domain Adaptation. *ICVSS, Sicily; GCPR, Heidelberg*, 2023
Urban Scene Semantic Segmentation with Coarse Annotation. *TRACE, Leuven*, 2022
Generalised Zero-shot Semantic Segmentation with Superpixels. *GCPR*, 2021

Scholarships, Achievements and Awards

Best Master's Thesis Award in Germany, GCPR 2021
Saarland Stipendium, funded by DAAD 2021
Saarbrücken Graduate School of Computer Science Scholarship 2019
Three Qualstars at Qualcomm India 2018
Ranked 6th of 185 students in Information Technology, IIIT Allahabad 2018
Institute Merit Scholarship, IIIT Allahabad (top 5% of class) 2017
43rd in Asia, WorldQuant Websim Challenge 2014
DST INSPIRE Scholarship (top 1% in India) 2012
Qualified for Indian National Mathematics Olympiad 2011